Build your own UAV

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Bosh/ENAC

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UAV





UAV





UAV





UAV

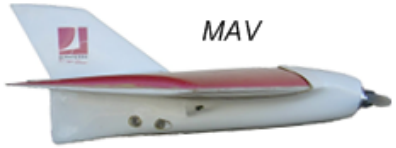




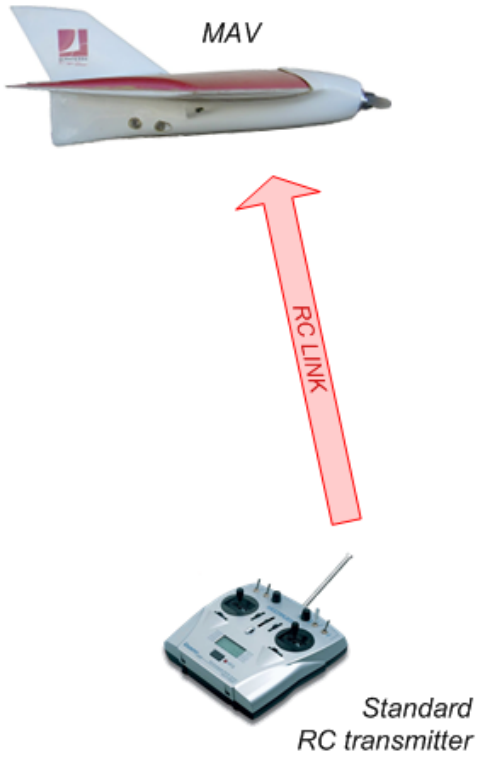
UAV



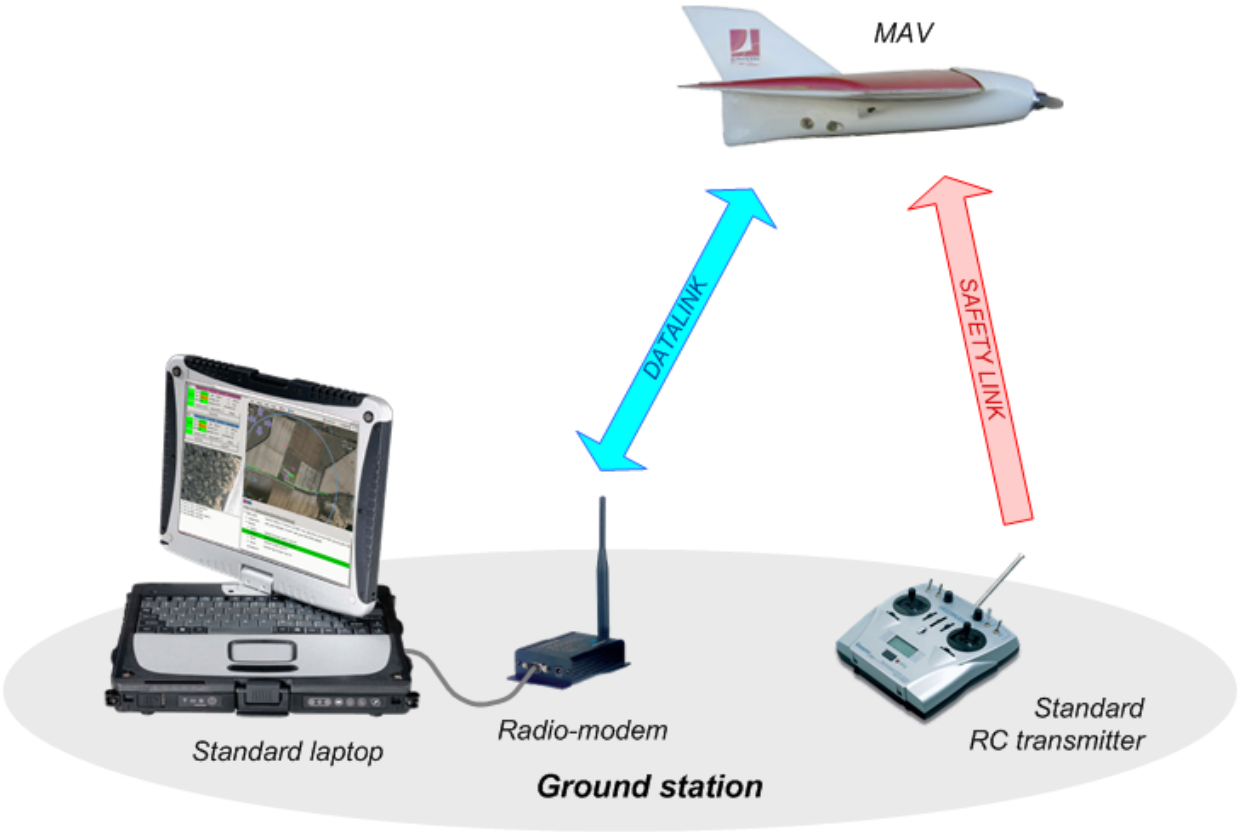
UAS



UAS



UAS

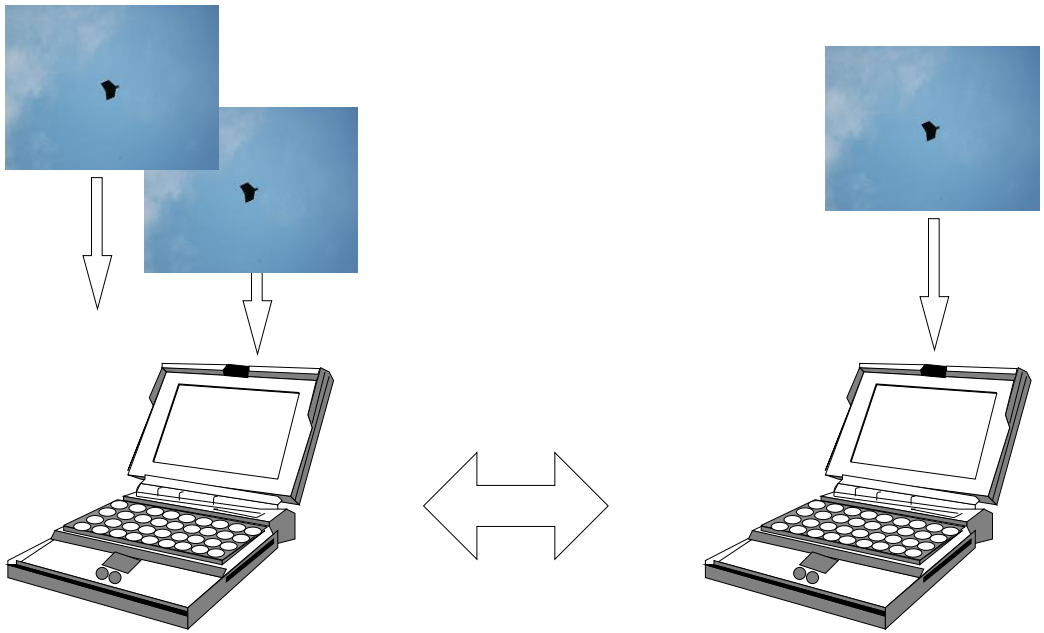

MAV
DATALINK
SAFETY LINK
Standard laptop
Radio-modem
Standard
RC transmitter
Ground station

UAS

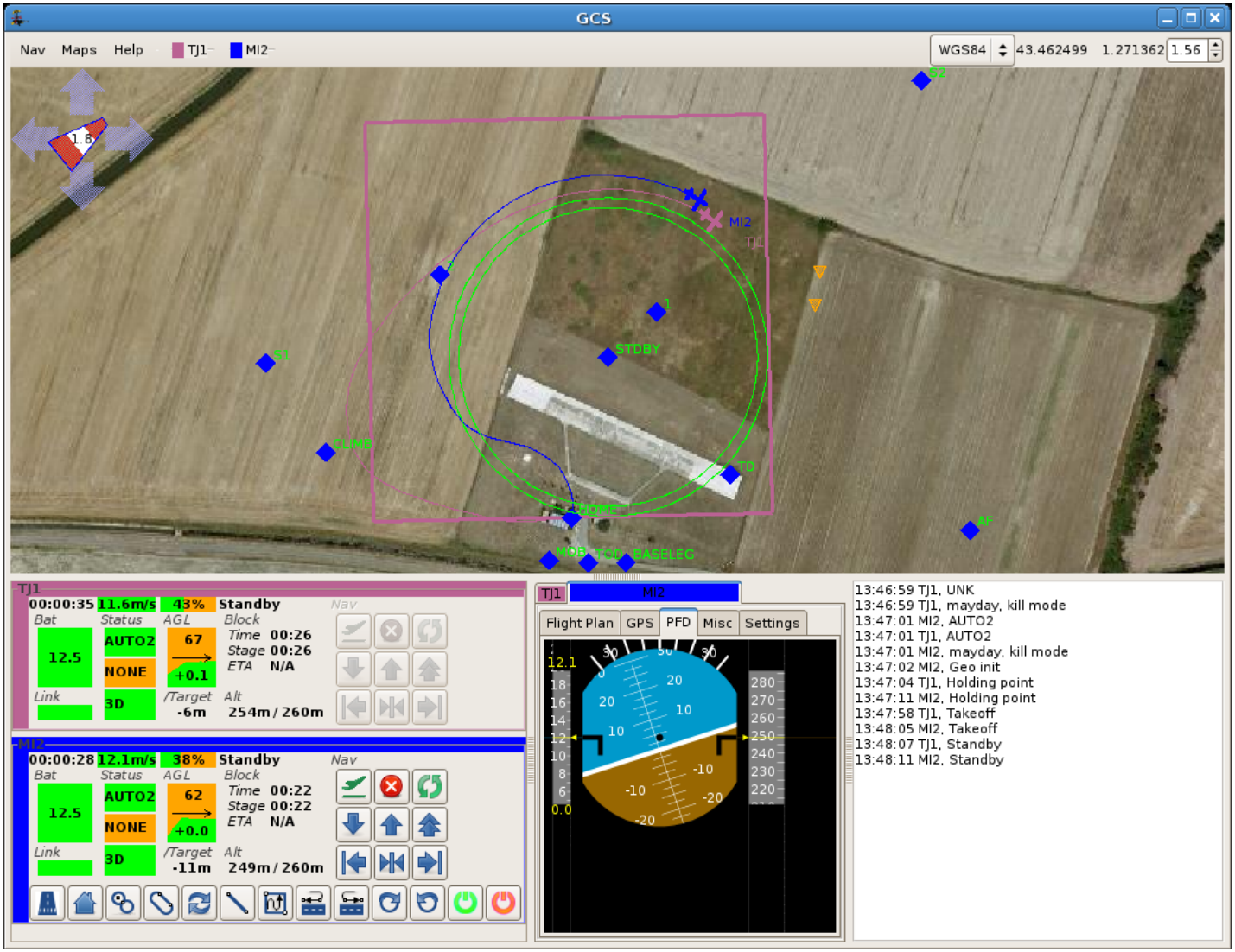

Processor
Sensors
Actuators
Propulsion
Payload
MAV
Preparation
Simulation
Tuning
Monitoring
Analysis
DATALINK
SAFETY LINK
Standard laptop
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Standard
RC transmitter
Ground station

Architecture

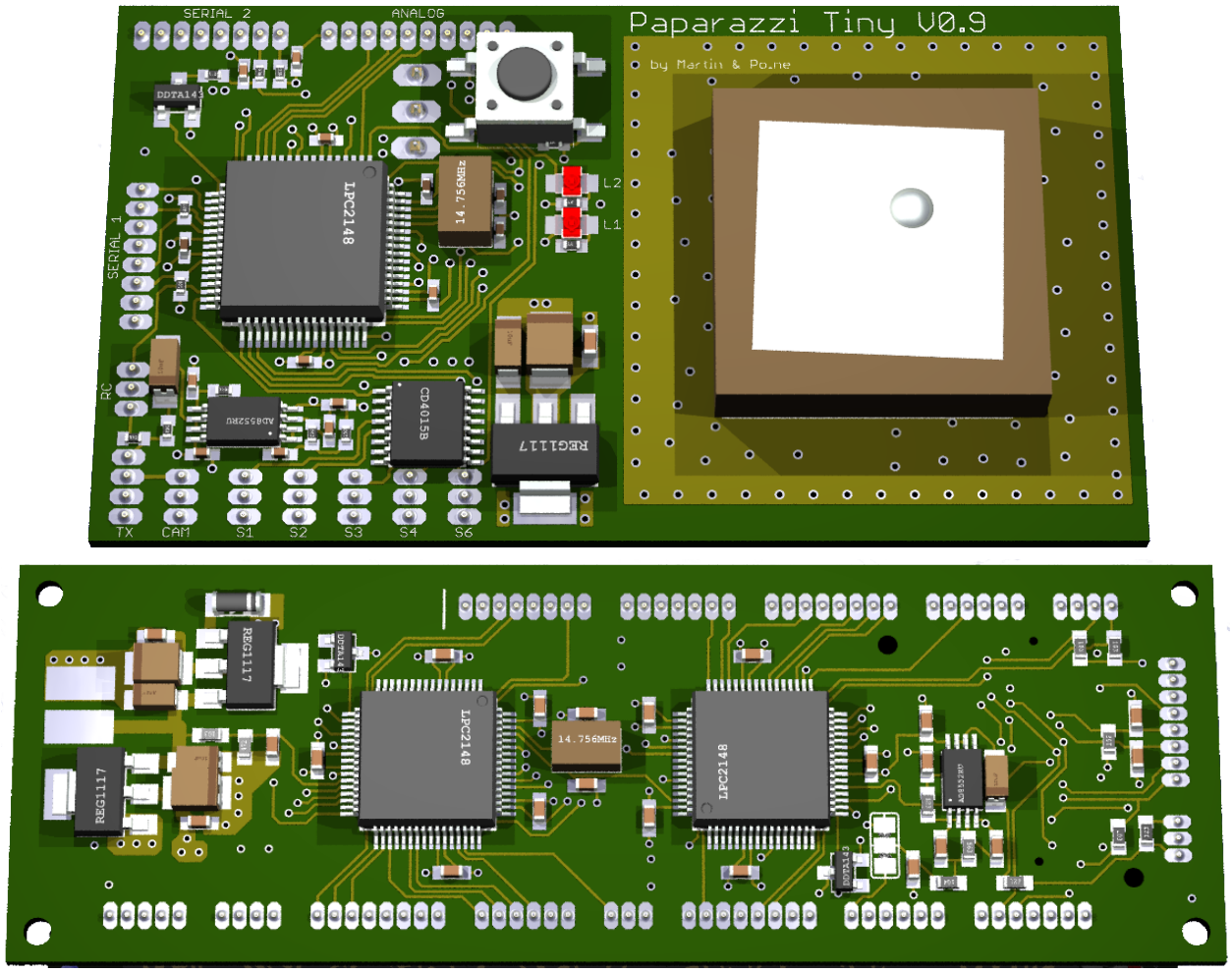
Multi-aircraft network enabled ground segment



Multi-UAV Human Machine Interface



Controller board



Video

- Real-time video downlink
- Footprint on map
- Network streaming
- Stitching
- Smoke detection





Video

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- Footprint on map
- Network streaming
- Stitching
- Smoke detection







MAV Competitions



- JMD03, Toulouse, France: 1st place with the Twinstar
- EMAV04, Braunschweig, Germany: 1st place with the Microjet
- JMD04, Toulouse, France: 1st place with the Microjet
- MAV05, Garmisch, Germany: 4 Paparazzi teams at the first 4 places
- EMAV06, Braunschweig, Germany : all the teams were equiped with Paparazzi
- MAV06, Sandestin, Florida: 2nd and 3rd places
- MAV07, Toulouse, France: 1st place (tie), 3rd, 4th and 5th places



Teaching/Reasearch





Forest fire detection





Meteorological surveying





Meteorological surveying





Conclusions



- COTS + Free Software enable affordable mini/micro UAVs



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- Wide fields of applications



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- COTS + Free Software enable affordable mini/micro UAVs
- Wide fields of applications
- Reglementation lagging



Acknowledments



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- Arnorld XXX



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- Michel Gorraz
- Anton Kochevar
- Christian Lindenberg
- Arnorld XXX
- Jeremy Tyler
- and all others...



Further Informations



- http://paparazzi.nongnu.org
- http://paparazzi.enac.fr
- irc://irc.freenode.net/#paparazzi